"TRUSTCHAIN AI: AI-POWERED DOCUMENT AUTHENTICATION, FRAUD DETECTION & BLOCKCHAIN-BASED DIGITAL CREDENTIAL VERIFICATION PLATFORM"

A MICRO PROJECT SYNOPSIS REPORT

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Dr. Babasaheb Ambedkar Technological University, Lonere.

Submitted By

1) Alok Sanjay Patil (Exam Seat No: 2354491243001)
2) Student Name 2 (Exam Seat No: 2354491243002)
3) Student Name 3 (Exam Seat No: 2354491243003)
4) Student Name 4 (Exam Seat No: 2354491243004)

of

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Under the Guidance of

Prof. Priti V. Sonawane

Department of AI & DS Engineering

Department of AI & DS Engineering
YSPM's Yashoda Technical Campus
Faculty of Engineering
Wadhe, Satara - 415011

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PROJECT ADDRESSING SUSTAINABLE DEVELOPMENT GOALS (SDGs)

TrustChain AI directly addresses the United Nations Sustainable Development Goals (SDGs) adopted under the 2030 Agenda. The platform aligns most significantly with SDG 4: Quality Education, SDG 9: Industry, Innovation and Infrastructure, SDG 16: Peace, Justice and Strong Institutions, and SDG 17: Partnerships for the Goals.

Under SDG 4: Quality Education, TrustChain AI addresses the systemic crisis of academic credential fraud, which undermines trust in legitimate qualifications worldwide. According to reports by the Higher Education Degree Datacheck (HEDD), over 36,000 fraudulent degree claims were detected between 2019 and 2022. The University Grants Commission (UGC) of India has repeatedly issued statutory warnings against fake degrees and unaccredited diploma mills. TrustChain AI combats this by establishing a decentralized, cryptographically unforgeable proof-of-issuance mechanism. When educational institutions register certificates, SHA-256 byte-level digests are immutably anchored to the Ethereum blockchain, enabling instant verification by employers and evaluators without slow, corruptible manual processes. This directly defends the value of legitimately earned academic qualifications and preserves merit-based educational equity.

Under SDG 9: Industry, Innovation and Infrastructure, the platform pioneers a convergent verification infrastructure combining computer vision (Error Level Analysis), optical character recognition, machine learning classification, and Ethereum smart contracts. By replacing archaic manual verifications requiring weeks with an automated, sub-three-second pipeline, TrustChain AI delivers a high-throughput digital public infrastructure suitable for both emerging and developed economies.

Under SDG 16: Peace, Justice and Strong Institutions, the platform fosters institutional transparency and anti-corruption through immutable audit trails. Every issuance, verification, and revocation event is permanently recorded in tamper-evident logs. The platform champions responsible AI principles: it provides calibrated risk tiers (Low Risk, Moderate Risk, Elevated Risk, Requires Manual Review) with explainable factor decompositions and disclaimers, supporting human adjudicators rather than imposing opaque automated verdicts.

Under SDG 17: Partnerships for the Goals, TrustChain AI establishes an open, interoperable standard. The smart contract registry supports multi-institutional onboarding, creating a collaborative digital trust network between universities, employers, and government agencies.

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1. ABSTRACT

TrustChain AI is an enterprise-grade hybrid AI and decentralized ledger verification platform engineered to eradicate document fraud, diploma mills, and credential forgery. The platform implements a fourteen-stage verification pipeline combining computer vision, layout-aware OCR, Ridge document classification, Error Level Analysis (ELA) for pixel-level tampering detection, cross-field logical validation, calibrated multi-factor risk scoring, and Ethereum blockchain cryptographic anchoring via a custom Solidity smart contract on Sepolia. The system incorporates a Retrieval-Augmented Generation (RAG) assistant grounded in verified university guidelines with anti-hallucination guardrails. Featuring a responsive React 19 frontend, asynchronous FastAPI backend, role-based access control, and an 8-section forensic dossier, the zero-PII blockchain architecture ensures complete privacy compliance while 74 automated tests validate system reliability.

2. INTRODUCTION

Credential forgery represents an escalating global challenge across academic, corporate, and governmental institutions. Modern image editing tools enable rogue actors to alter certificates with pixel-level precision. Current manual verification processes are slow, expensive, and centralized repositories present single points of failure. TrustChain AI unifies computer vision, decentralized blockchain ledgers, and RAG natural language intelligence into a cohesive, high-throughput verification ecosystem that evaluates intrinsic document forensics and matches raw cryptographic hash digests against the Ethereum Sepolia network within seconds.

3. PROBLEM STATEMENT

The current landscape of document verification is compromised by manual inspection inability to detect sophisticated digital forgeries, the absence of a unified, tamper-proof credential registry, and the lack of automated, explainable, and privacy-preserving verification mechanisms that operate at scale. There exists a critical need for an integrated platform combining AI forensic analysis, blockchain cryptographic anchoring, and grounded natural language intelligence to provide evidence-based, calibrated document authentication in real time.

4. OBJECTIVES

• Develop a fourteen-stage automated document verification pipeline integrating CV, OCR, and ELA tampering forensics.
• Deploy a zero-PII Solidity smart contract registry on Ethereum Sepolia for tamper-proof credential notarization.
• Implement a grounded RAG intelligence system providing citation-backed explanations with anti-hallucination guardrails.
• Build a responsive React 19 web application with an 8-section forensic dossier and responsible AI risk scoring.

5. SCOPE OF PROJECT

TrustChain AI covers six credential categories: academic degrees, marksheets, corporate internship certificates, training certifications, experience letters, and educational IDs. The platform operates on Ethereum Sepolia testnet with a zero-PII model and sub-3-second CPU inference.

6. LITERATURE REVIEW

[1] Kumar et al. (2023): Proposed Ethereum smart contract certificate verification with IPFS, but lacked content-level AI forensics.
[2] Chen et al. (2023): Comprehensive survey on deep learning document fraud detection, identifying need for localized ELA and blockchain linkage.
[3] Lewis et al. (2020): Established RAG architecture combining parametric LLMs with dense vector retrieval to eliminate hallucinations.

7. BLOCK DIAGRAM

Ingestion ➔ 300 DPI Preprocessing ➔ Layout OCR ➔ Ridge Classifier ➔ ELA Forensics ➔ Cross-Field Validation ➔ Risk Scoring (0-100) ➔ Ethereum Sepolia Lookup ➔ RAG Vector Grounding ➔ 8-Section Dossier & PDF Export

8. METHODOLOGY & CONCLUSION

Developed across four modular phases: Application Foundation, AI/ML Forensics, RAG & Blockchain Integration, and End-to-End Hardening with 74 tests. TrustChain AI demonstrates a practical, scalable, and ethically responsible solution to real-world credential fraud.

9. REFERENCES

[1] S. Kumar et al., J. King Saud Univ. Comput. Inf. Sci., 2023. | [2] M. Chen et al., IEEE Access, 2023. | [3] P. Lewis et al., NeurIPS, 2020.